

gemstone-py Plan3 Feature Map

Mapping improvement streams to modules, examples, docs, and release checks

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Plan3 Feature Map

This page maps the plan3 improvement streams to the code, examples, and docs that now carry each feature. It is a quick orientation page for release review, new contributors, and the VS Code workbench docs view.

The same map is available from the command line:

```
gemstone-examples plan3-map
python -m examples.cookbook.plan3_feature_map
```

Status

Stream	Status	Main Surface
1. Pool + health	Landed	<code>GemStoneSessionPool</code> , <code>AsyncSessionPool</code> , pool-backed FastAPI/Litestar dependencies
2. Streaming results	Landed	<code>GSCollection.search_iter(...)</code> , <code>GSCollection.iter(...)</code> , <code>Query.iter(...)</code> , async query iteration
3. Typed codegen	Landed	<code>gemstone-codegen</code> , generated wrappers, async wrappers, <code>.pyi</code> stubs, drift checks
4. Observability	Landed	tracing, metrics, slow-operation logging, pool/session observations
5. Migrations	Landed	module migrations, rollback, dry-run, class diff, schema fingerprinting
6. Inspect/debug	Landed	<code>session.inspect(...)</code> , <code>session.dump(...)</code> , <code>session.describe_class(...)</code> , <code>gemstone-inspect</code>
7. Framework adapters	Landed	<code>web_core</code> , Flask/Django adapters, FastAPI/Litestar async adapters
8. Examples	Landed	quickstart, examples guide, cookbook map, workbench example runner
9. Performance docs	Landed	benchmark CLI, compare/register helpers, committed baseline docs
10. Native wheels	Landed	PyO3 native package, stable-ABI wheel workflow, release checks
11. Bootstrap	Landed	

Stream	Status	Main Surface
		packaged GemStone-side <code>init.st</code> , <code>gemstone-bootstrap</code> , setup docs

Stream Details

Stream	Modules	Examples	Docs
Pool + health	<code>gemstone_py.session_providers</code> , <code>gemstone_py.aio.pool</code>	<code>examples/</code> <code>fastapi/</code> , <code>examples/</code> <code>litestar/</code>	User Manual , Cookbook
Streaming results	<code>gemstone_py.gsquery</code> , <code>gemstone_py.aio.gsquery</code>	<code>examples/</code> <code>async_features/</code>	Examples Guide , Performance
Typed codegen	<code>gemstone_py.codegen</code>	<code>examples/</code> <code>typed_access/</code> <code>codegen_demo/</code>	Type-Safe Smalltalk Codegen
Observability	<code>gemstone_py.observability</code>	<code>examples/</code> <code>fastapi/</code> , <code>examples/</code> <code>litestar/</code>	Observability
Migrations	<code>gemstone_py.migrations</code>	<code>examples/</code> <code>persistence/</code> <code>migrations/</code>	Cookbook , User Manual
Inspect/ debug	<code>gemstone_py.inspection</code> , <code>GemStoneSession.inspect</code> , <code>GemStoneSession.dump</code> , <code>GemStoneSession.describe_class</code>	<code>examples/</code> <code>cookbook/</code>	User Manual
Framework adapters	<code>gemstone_py.web_core</code> , <code>gemstone_py.frameworks</code> , <code>gemstone_py.aio.fastapi</code> , <code>gemstone_py.aio.litestar</code>	<code>examples/</code> <code>fastapi/</code> , <code>examples/</code> <code>litestar/</code> , <code>examples/django/</code>	Framework Adapters
Examples	<code>gemstone_py.cli</code>	<code>examples/</code> <code>quickstart.py</code> , <code>examples/</code>	Examples Guide , examples/ README.md

Stream	Modules	Examples	Docs
		webstack/, examples/ cookbook/	
Performance docs	gemstone_py.benchmarks , gemstone_py.benchmark_compare	examples/ native_backend/	Performance
Native wheels	gemstone_py.native , gemstone-py-native/	examples/ native_backend/	README , Release Checklist
Bootstrap	gemstone_py.bootstrap , gemstone_py/_gemstone_side/	examples/ quickstart.py , examples/ cookbook/	Setup Guide

Verification Gates

Before treating the plan3 batch as release-ready, run the local checks:

```
.venv/bin/python -m ruff check gemstone_py tests
.venv/bin/python -m mypy
.venv/bin/python -m pytest -q
./scripts/check_codegen.sh
```

Then run the live GemStone checks on a configured stone:

```
GS_RUN_LIVE=1 ./scripts/run_live_checks.sh
```

For release packaging, validate the native package workflow and the workbench extension release checks:

```
./scripts/run_native_checks.sh
cd vscode-gemstone-py-workbench
npm run release:preflight
npm run release:verify-vsix
```

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